

Safety Scoring Model: Technical Specification

1. Objective

The purpose is to compute a safety score (0–100) for a given route based on nearby crime data. The score reflects both the severity and recency of incidents, normalized by route length.

2. Spatial Filtering

Crimes are included if they fall within a fixed radius of any point along the route.
Threshold distance: $d \leq 0.18$ km

3. Distance Calculation (Haversine)

The distance between two geographic points is computed using the Haversine formula:

$$d = 2R * \text{atan2}(\sqrt{a}, \sqrt{1 - a})$$

$$a = \sin^2(\Delta\phi/2) + \cos(\phi_1)\cos(\phi_2)\sin^2(\Delta\lambda/2)$$

Where $R = 6371$ km.

4. Route Length Normalization

Total route distance:
 $D = \text{sum of segment distances}$

Normalized distance:
 $D_{\text{safe}} = \max(D, 1.0)$

5. Crime Weighting

Each crime is assigned to a severity weight (w_{type}).

- "HOMICIDE": 5,
- "SEX ABUSE": 4,
- "ASSAULT WITH DANGEROUS WEAPON": 4,
- "ROBBERY": 3,
- "MOTOR VEHICLE THEFT": 3,
- "THEFT/AUTO": 2,
- "THEFT/OTHER": 2,

- ▽ Temporal decay is applied:
 $w_{\text{year}} = 0.8^{\text{age}}$
 $\text{age} = \text{currentYear} - \text{crimeYear}$

6. Aggregate Risk

Total weighted crime score:

$$W = \sum (w_{\text{type}} \times w_{\text{year}})$$

Normalized risk:

$$R = W / D_{\text{safe}}$$

7. Final Safety Score

The safety score is computed using exponential decay:

$$\text{Score} = 100 \times \exp(-\lambda R)$$

$$\lambda = 0.0004$$

Score is bounded between 0 and 100.

8. Assumptions

- Risk contributions are additive
- Crime severity can be approximated using fixed weights
- Risk decays exponentially over time
- Risk is localized to a fixed spatial threshold
- Safety is inversely proportional to crime density per km

9. Limitations

- Weights are manually assigned
- Assumes linear contribution of events

10. Future Work

- Learn weights using regression or optimization
- Incorporate real-time incident data
- Use safety score as edge weights
- Transition from heuristic model to data-driven model